

Education

University of Utah

B.S. Computer Science

Salt Lake City, UT

August 2020 – May 2025

Relevant Coursework:

Computer Systems, Machine Learning, Computer Graphics, Algorithms, Software Practice I & II, Database Systems, Computer Networks, Foundations of Data Analysis, Models of Computation, and Linear Algebra.

Skills

Languages:

Python, TypeScript, C++ , SQL, Java

Software:

Pandas, NumPy, React, Node.js, AWS, Azure, Docker, Next.js, .NET, REST, Django, MongoDB, Linux

Experience

L3Harris Technologies

Associate Software Engineer

Dallas, TX

June 2025 – Present

• Built microservices and automation tooling from scratch for high-frequency defense sensor data ingestion, owning the full lifecycle from architecture through deployment and production support with no prior template to follow.

• Identified and closed gaps in CI/CD and testing coverage on my own initiative, shipping pipeline automation that reduced manual effort and improved release reliability across engineering teams.

• Collaborated directly with hardware and firmware engineers to translate ambiguous system requirements into working software, operating without a product manager or detailed spec.

Doxy.me

Software Engineer

Charleston, SC

August 2024 – May 2025

• Built a real-time mental health inference system from scratch for Doxy.me’s telehealth platform, fine-tuning Llama models to predict depression risk and mood indicators from tone, voice, and facial cues over live WebRTC sessions — owned the problem end to end from model selection through SageMaker and Fargate deployment.

• Improved diagnostic accuracy by designing and closing a reinforcement learning feedback loop over physician-validated outcomes, shipping iterative improvements to production without a dedicated ML research team.

• Shortened the gap between model output and clinician action by building TypeScript frontend components embedded in the video consultation interface, surfacing AI-generated insights directly at the point of care.

University of Utah

Undergraduate Research Assistant – FuTURES Lab

Salt Lake City, UT

May 2024 – Jun 2025

• Analyzed and optimized large-scale datasets, enhancing software configuration testing with gcov and CMake, increasing code coverage by 30% on real-world APIs (Libpng, PyTorch).

• Expanded OSS-Fuzz testing coverage using compile-time options to improve the robustness of full-stack libraries.

HEXstream

Software Engineering Intern

Chicago, IL

May 2022 – Aug 2022

• Engineered backend ETL pipelines integrating data from 25+ enterprise sources into Azure SQL and Azure Data Lake.

• Developed automated workflows for ingestion, cleansing, and aggregation, supporting distributed analytics systems.

Projects

Telehealth Mental Health Diagnosis Tool (Doxy.me):

Built an AI system that analyzes visual, speech, and tonal cues from telehealth sessions using PyTorch and Llama, delivering structured mental health insights to treating physicians via SageMaker and Fargate inference pipelines. Improved diagnostic accuracy using reinforcement learning and WebRTC video integration.

University Course Planning/Review Platform (Class Best):

Built a multi-service data platform enabling student schedule matching, course overlap detection, and behavioral insights. Developed ETL pipelines in Python to process thousands of course records, user preferences, and time-series activity logs. Built a responsive frontend in React + Tailwind with interactive dashboards, heatmaps, and calendar visualizations.

Configuration Fuzzer:

Developed a configuration fuzzing tool for OSS-Fuzz, automating build generation to identify critical compile-time configurations and improve code coverage, uncovering previously untested code paths and increasing bug detection effectiveness.